

Modular Design Architecture of a HPEC System

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Abstract—The current High Performance Embedded Computing (HPEC) platform is an embedded server based on VITA VPX Standards. It is a 3/4-ATR system with 4-slots based on VPX VITA 46 standards in 3U form factor. It has two independent SBCs based on Intel Xeon-D and Xeon-E series, a GPGPU module based on Nvidia and an IO card for MIL-1553 and ARINC applications. The design of such a complex system meeting the performance, size, thermal and EMI/EMC challenges is a herculean task in its own right. In addition to these common challenges in such a system, the current paper discusses about efforts to make the system modular to ensure faster time to market for different variants. In addition, the high TFLOPS performance of such systems make them ideal for AI applications.

Keywords—High Performance Embedded Computing (HPEC) systems, Modular Computing, GPGPU, Computing Server

I. INTRODUCTION

In view of the current advancement in technology and dynamic fluctuations in module availability, it is often necessary to redesign the entire system due to changes in the sub-modules used. The current design proposes to address these changes in an effective manner and faster time to market. The high-speed signals provide an integrated system with all the modules logically connected to form a single coherent system. The modular nature of the design allows for easy maintenance of the system. Each sub-module further designed in such a way that the sub-modules conform to the industry standard form factor and hence can be upgraded without any change in the system design. This allows swapping of any module due to obsolescence or different performance parameter.

The modular server system provides a compact, modular, efficient, and powerful multi-server system that is highly reliable and easy to maintain. The high density of the modular server system reduces real estate costs, and the modular nature of its field replaceable units allows repairs and replacements to be performed quickly and easily.

II. DESIGN

The current system is capable of data link processing responsible for all core functions including data acquisition, transformation, correlation and storage. This modular system, as shown in Fig.1, consists of the following modules:

- a) Configurable Compact Computing Hardware-1 (100)
- b) Configurable Compact Computing Hardware-2 (300)
- c) Configurable Graphics Processing Hardware (500)
- d) Configurable Communication Hardware (400)
- e) 3U VPX Power Supply (700)
- f) Configurable Data Storage Hardware (200)
- g) Customized Motherboard (600)

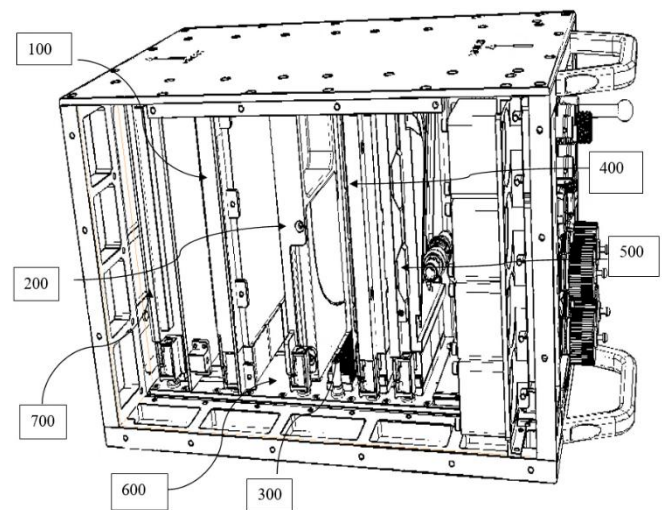


Fig.1 HPEC system representation

These modules are designed to meet industry-approved standards and are arranged based on thermal and structural analysis of the system for improved thermal performance. These modules are logically connected with each other through high-speed link existing on the motherboard. Data storage module has a standard provision for data storage and retrieval of information. Industry standard connectors are used to bring out all the important interfaces onto the front panel.

The Computing Hardware-1 is the system master. The high-speed PCIe protocol is used to interface the various modules with the Computing Hardware-1. A brief description about each sub module is discussed in the following topics.

A. Motherboard

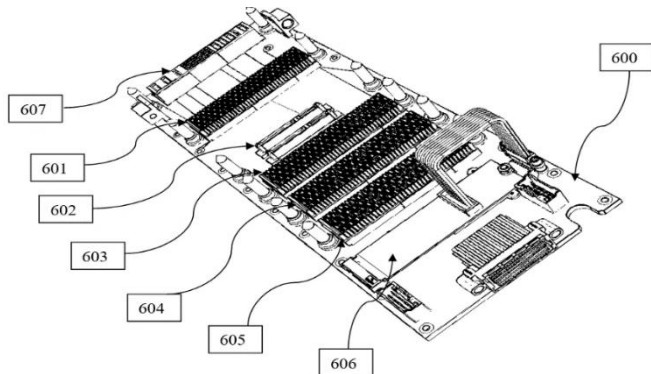


Fig.2 Customized Motherboard

The customized motherboard with standardized connectors and standard interfaces used for interconnect architecture, allows for future upgrades of all the modules. The figure shows the media for communication (Motherboard) of the modular computing platform system. This is a 3U VPX compliant motherboard with 4 VPX VITA 46 slots (601,603,604,605), one VPX VITA 62 power supply slot (607) and a SATA cassette slot (602) having two SSD hard disk for storage. These modules are interconnected on a motherboard to meet the needs of both tree and mesh topologies. There is a data interchange switch (609) for high-speed interconnect to convert the x4 lane to four x1 lane for the ARINC, mil-1553, computing hardware-2 and m.2 connector (606). The m.2 connector is used to convert PCIe protocol to the HDMI for high-resolution display. mSATA (608) mounted on the rear side of the motherboard to meet the storage requirements for the system. The micro-controller on mother board is used for soft shutdown, monitoring the system temperature and controls the fans RPM according to the surface temperature of the system.

Some of the important features of the Motherboard are as follows:

1. VPX Slots: 5 nos (Including Power Supply Slot)
2. PCIe Signals:
 - a. x16 Gen 3
 - b. 4 x1 Gen 2.0
3. Gigabit Ethernet : 2 nos
4. SATA 3.0 (6 Gbps) : 3 nos
 - a. USB :
 - b. 2 nos of USB 3.0
 - c. 4 nos of USB 2.0
5. HDMI : 2 nos

6. MIL 1553 bus: 1 nos
7. ARINC 429 interface : 8 channels of Rx and two channels of Tx
8. Serial Ports : 2nos

B. Configurable Compact Computing Module-I & II

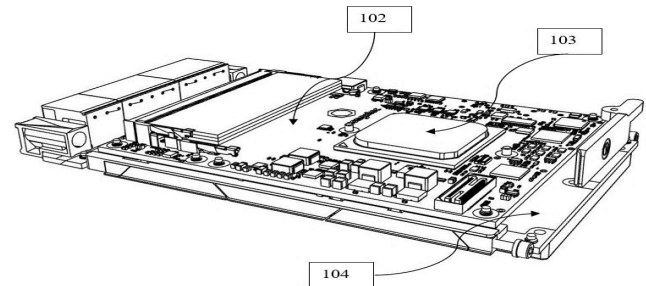


Fig.3 Configurable computing module

The computing hardware is of standard form factor, which holds an add-on industry standard system on module connected through industry standard interface on carrier card, which can be upgraded without change in design. These SOM (System-on-Module) modules are mounted on the carrier cards through industry standard connectors and modules can be updated without change in design. The modules act as system controllers and run independently. They also include high-speed signals, serial peripheral interfaces and transfer protocols for different applications. These computing hardware have customized conduction-cooled heat sink designed for thermal dissipation.

The critical challenge is the electro-mechanical design to ensure the SBC fits in the 3U form factor with an add-on module. This required innovative product design solution.

The important features of the current Computing Hardware modules are:

Sr.No	Parameters	Description
1	Form Factor	3U Open VPX architecture. (Conduction Cooled, 1" Width)
2	Processor	Intel Xeon D-1559
3	PCIe	x16 Gen3 x4 Gen 2
4	No.of Cores	12
5	DRAM	32 GB DDR4
6	Storage Memory	1 TB - SSD (Software erasable feature)
7	USB Ports	USB 3.0/ USB 2.0 - 4 Nos.
8	Gb Ethernet Port	2x 1000 T Nos. (IPv6 capable)
9	Display Port	DVI-I /SDI
10	SATA Ports	2 Nos. (6 Gb/s)
11	Operating System	RHEL 8.3

Table 1: Computing Hardware module Specification

C. Graphics Processing Module

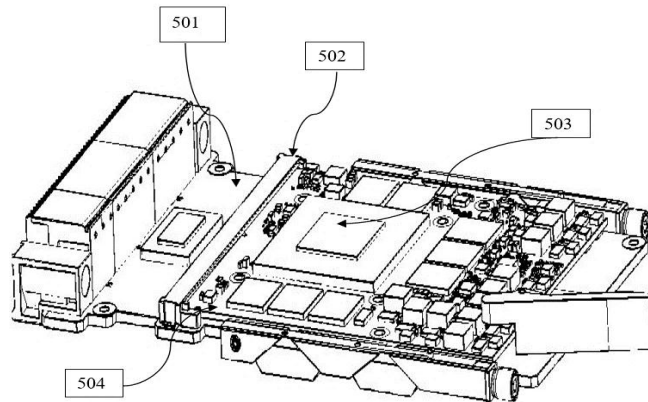


Fig.4 Graphics Processing Module

The Fig.4 shows a graphics processing hardware of the modular computing platform system. The graphics processing Hardware (500) is a General-Purpose Graphic Processing Unit (GPGPU) having a graphic card (504) with a pluggable high performance graphics mobile PCI express mezzanine (MXM) connector (502). This graphic card is mounted on a base carrier card (501) designed in VPX VITA 46 standard in 3U form factor. This module can be updated without change in design, by replacing the graphic modules available in the industry.

In the present system, the GPGPU is based on NVIDIA A4000 [3], the powerful single slot GPGPU solution offering high performance real-time ray tracing, AI-accelerated compute, and professional graphics rendering. Building upon the major enhancements from the Turing GPU, the NVIDIA Ampere architecture enhances ray tracing operations, tensor matrix operations, and concurrent executions of FP32 and INT32 operations. A4000 supports PCI Express Gen 4, which provides double the bandwidth of PCIe Gen 3, improving data-transfer speeds from CPU memory for data-intensive tasks like AI and data science.

The technical specifications of the module are as follows:

1. Graphic Core : NVIDIA RTX™ A4500 GPU
2. Memory : 16GB GDDR6 memory
3. CUDA Cores : 5888 CUDA cores
4. GPGPU Computing : 17.8 TFLOPS peak FP32 performance
5. Tensor Cores: 160 Tensor Cores (performance 153.4 TFLOPS) + 40 RT tensor cores (performance 37.4 TFLOPS).
6. Compute API : CUDA Compute version 8 and above, OpenCL™ 1.2
7. Graphics API : DirectX® 12, OpenGL 4.6
8. Display Outputs : 4K at 120Hz or 8K at 60Hz with 10-bit color depth

The widely used previous generation Quadro RTX 4000 is used for comparison of performance, power and cost.

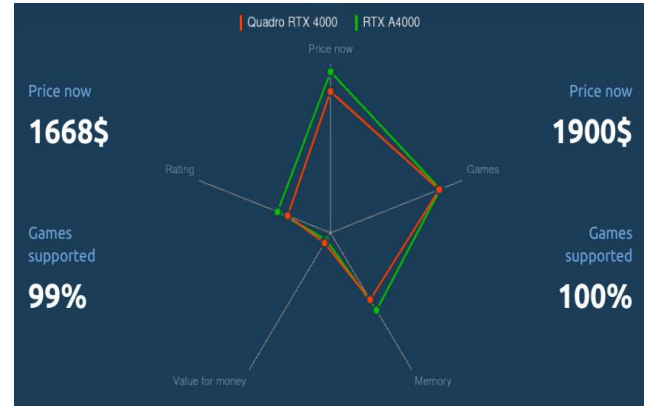


Fig.5 Price and Performance Comparison Matrix

The Performance comparison with various benchmark softwares is as shown Fig.6:

GPU 1: NVIDIA RTX A4000 Mobile
GPU 2: NVIDIA Quadro RTX 4000

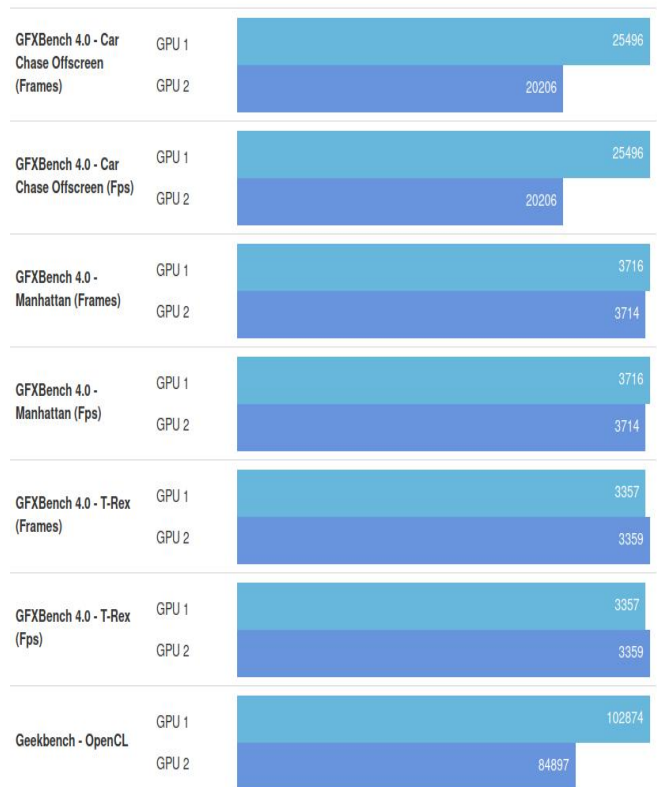


Fig.6 Benchmark Softwares Performance Comparison [7]

From the technical specifications, we can see that the GPGPU module has very high processing capabilities. This makes it very much suitable for applications like AI and high-end graphics processing including medical imaging. This high computing power comes at very high power consumption and thermal dissipations that needs to be addressed. The current module consumes 120 W of power

at peak performance. A customized conduction cooled heat sink for thermal dissipation has been designed in house after several iterations with respect to the material used and heat pipes design.

D. Configurable Communication Module

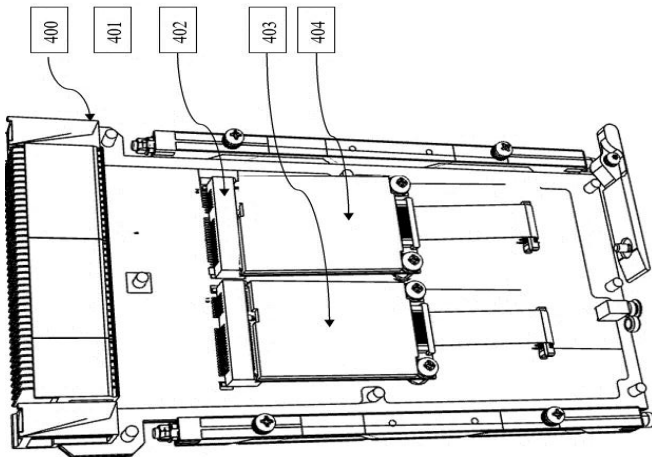


Fig.7 Configurable Communication Module

The Fig.7 shows the communication hardware of the modular computing platform system for a motherboard. The communication hardware consists of small form factor Mini-PCIE connectors (402) mounted on the base carrier board (401) to provide a cost effective, lightweight, small size, rugged, and reliable bus interfaces. This Module supports MIL-STD-1553(403) and ARINC 429 (404) military, COTS grade cards to meet data conversion and data interfaces. ARINC 429 protocol ensure the accurate and reliable transfer of flight-critical data. This module can be easily updated without change in the design by upgrading the add-on cards on Mini-PCIE connector. The entire module also can be easily changed to handle other communication protocols by changing the industry standard add-on modules.

E. Configurable Data Storage Module

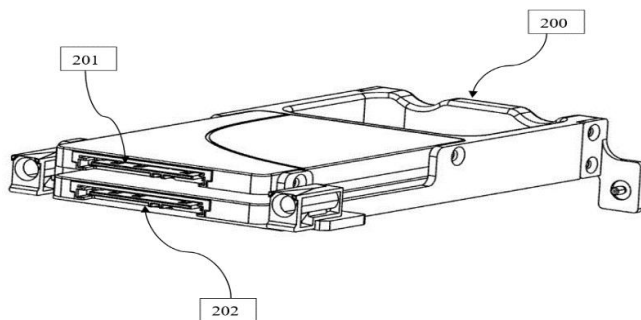


Fig.8 Configurable Data Storage Module

The Fig.8 shows the storage module of the modular

computing platform system for a motherboard. The storage module (200) with two industrial standard SSDs (201 & 202) have been packed in customized mechanical enclosure mounted on the motherboard for computing hardware-1 and computing hardware-2 respectively for storage of data and retrieval of information. The storage module can be further upgraded based on storage capacity requirement and technology solutions from vendors across the industry.

F. 3U VPX Power Supply

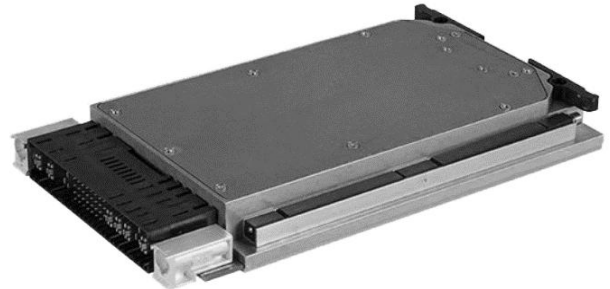


Fig.9 VPX power Supply Module

The present invention system further includes a Power supply hardware, Fig.9, which is a 300W industry standard VITA 62 compliant power supply module with wide input range and has a HOLDUP time up to 50mSec. It has special features which includes indefinite short circuit protection, over-voltage shutdown with auto-recovery, reverse battery protection and over temperature shutdown with auto-recovery. It's also designed to meet Design to meet with MIL-STD 461F (5 μ H LISN): CE101, CE102, CS101, CS114, CS115, CS116.

III. DESIGN ANALYSIS

The development of complex high performance computing systems comes with it's own challenges. The use of high-speed signals require Signal integrity Analysis. The high power consumption nature requires proper power Integrity analysis. It also leads to excessive heat dissipation. Hence proper thermal load analysis of all the modules is very much required. The small form factor of the present day systems adds to these challenges. Some of the design improvements after these analyses are discussed in the following sections.

A. Signal Integrity Analysis

Signal Integrity analysis helps us to understand if the signal eye opening is sufficient as per the protocol requirements[1]. Generally, SI simulation considers the signals, which are part of same interface. 2D or 3D SI simulation softwares will not consider adjacent switching signals and unrelated signals for analysis. In the image Fig.10, we see one of the PCIe Gen3 signal not meeting the eye height requirement.

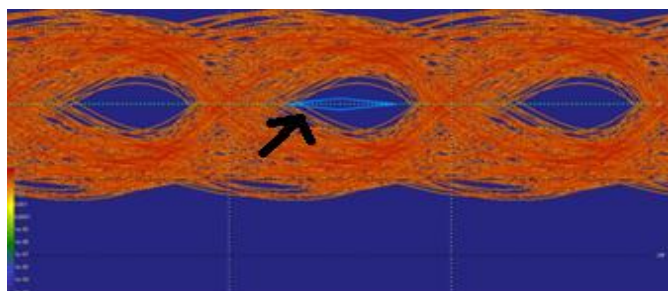


Fig.10 Eye opening not meeting the specification

The signal rerouting was done and SI analysis repeated. The improved SI analysis image is as shown Fig.11 :

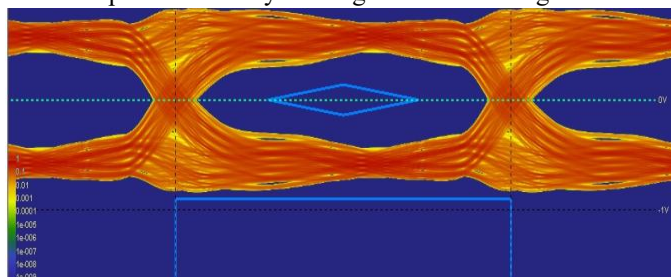


Fig.11 Eye opening meeting the specification

B. Power Integrity

Power Integrity analysis is done with respect to

- Voltage drop
- Ripple analysis
- Current density
- Transient load requirement

Computing boards require more number of power rails. Free space in the signal layer may be used for power planes of low current power rails, but care to be taken to ensure that the impedance of adjacent signal layers are not affected. In addition, when power planes for the same power rails are put on different layers, adequate stitching vias needs to be ensured [2]. Some of the core power rails require high current at very low voltage of less than 1V. Such extremely low voltages decrease the margin for voltage drop across the board to the order of 50 mV. PI analysis ensures adequacy of plane area & shapes across the board including signal layers. A good power plane design should reduce the current density. As per IPC-222, for permissible 10°C rise in temperature, the current density should not exceed 125mA/sq. mil for external layers & 62.5mA/sq. mil for internal layers[4].

The image Fig.12, shows the 12V power distribution across the different slots in the motherboard. It shows the current requirement for each slot and expected voltage at the slot after the voltage drop. We can see that the drop increases as we move further from the power source. While the drop cannot be completely eliminated, it has to be maintained within the acceptable tolerances.

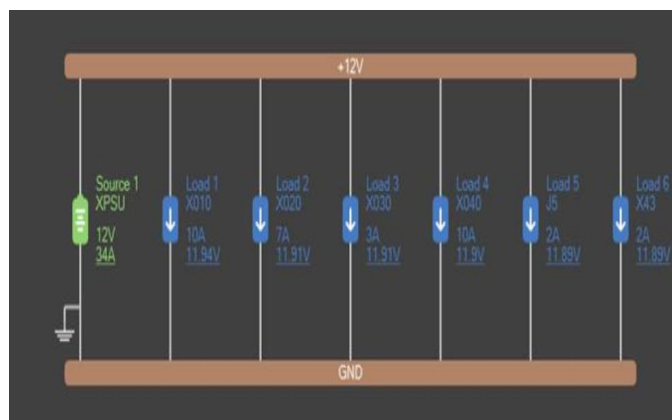


Fig.12 12V Current Consumption distribution

The below two images show the airflow with and without radius added for fin and with fin cut.

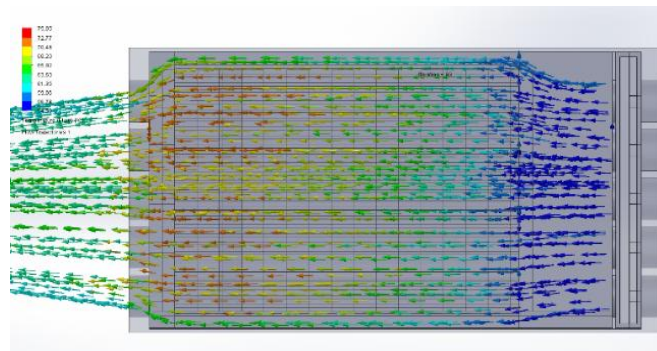


Fig.13 Airflow without radius fin added

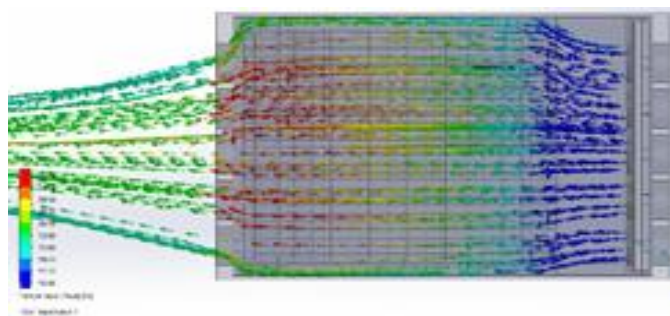


Fig.14 Airflow with radius fin added

From the above images Fig.13 and Fig.14, we see that , at 55°C ambient, the exhaust air at the other end of the chassis is almost 2°C lesser.

Similarly, the module level simulations ,with aluminum and copper based heat sink showed that copper based heatsink is better but the weight is more which will affect the vibration and shock requirements. The Thermal simulations are as shown in figures Fig.15 and Fig.16:

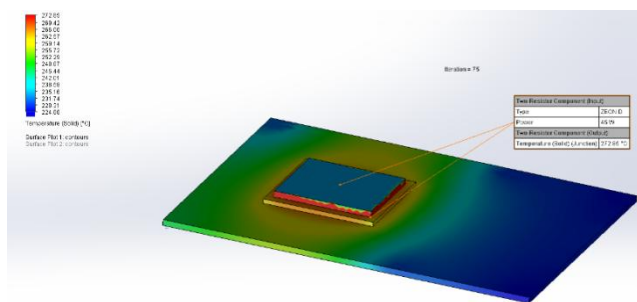


Fig.15 Temperature Gradient with Aluminum Heat Cover

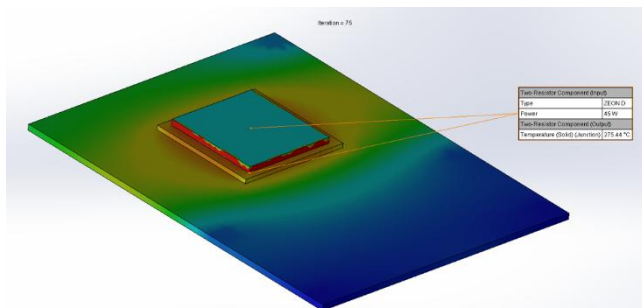


Fig.16 Temperature Gradient with Copper Heat Cover

IV. CONCLUSION

A High Performance Embedded Computing (HPEC) system has many parallel processing and distributed computing features, which offers superior overall performance compared to traditional computing platforms. These systems are the core of modern AI-computing, Image processing applications in Radars and Sonars, Communication backbone networks and other such computational intensive systems. However, these systems come with their own challenges due to the fast pace of electronics development, ever reducing size with increased power budget. All these need innovative solutions in product design and thermal management. The number of design iteration, analysis and simulations are also more. The cycle time between the design upgrades also reduces the development time available. Modular design helps in overcoming these challenges. In this paper, we have addressed the various modules involved in such systems and the design process.

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